GG-B-675D
September 23, 1976
SUPERSEDING
Int. Fed. Spec. GG-B-00675C(GSA-FSS)
October 7, 1974, and
Fed. Spec. GG-B-675A
February 19, 1964

FEDERAL SPECIFICATION

BREATHING APPARATUS, SELF-CONTAINED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers self-contained breathing apparatus for respiratory protection in oxygen-deficient or toxic atmospheres containing vapors or gases. Protection is not provided against poisoning by skin absorption.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Class and style. Self-contained breathing apparatus furnished under this specification shall be the one-half (1/2) hour duration open circuit design. They shall be of the class and style indicated in table I.

1.2.2 Definitive part number (DPN). Breathing apparatus covered by this specification shall be identified by a DPN. This part number is intended for cataloging and ordering purposes only. The DPN shall consist of the Federal Specification symbol plus the specification part number (SPN) (see table I). The DPN shall be written as indicated below (see 6.2).

DPN----------GGB675 - XX
Federal Specification Symbol----------
SPN (see table I)----------

TABLE I. SPN identification 1/

SPN	Class 1	Class 2
1B	Style B	-----
1D	Style D	-----
2B	-----	Style B
2D	-----	Style D

1/ Class and style identification:

Class 1 - Compressed air.
Class 2 - Compressed oxygen.
Style B - Demand type with audible warning device.
Style D - Pressure demand type with audible warning device.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

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Federal Specification:

PPP-B-636 - Box, Shipping, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 191/4108 - Strength and Elongation, Breaking; Textile Webbing, Tape and Braided Items.
Fed. Test Method Std. No. 191/5100 - Textile Test Methods/Strength and Elongation, Breaking of Woven Cloth; Grab Method.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Laws and Regulations:

30 CFR 11 - Respiratory Protective Devices; Tests for Permissibility; Fees.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof).

Military Specification:

MIL-P-116 - Preservation, Methods of.

Military Standards:

MIL-STD-101 - Color Code for Pipelines and for Compressed Gas Cylinders.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Air Force Manuals:

AFM 71-4 - Packaging and Handling of Dangerous Materials for Transportation by Military Aircraft.

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

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Uniform Classification Committee Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza., Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Approval. Self-contained breathing apparatus furnished under this specification shall comply with the requirements specified in 30 CFR 11. The breathing apparatus shall bear approval plates and markings for permissable apparatus and for approved supplemental parts as provided in 30 CFR 11 subpart D.

3.1.1 Certification. The supplier shall certify in writing to the contracting officer or his representative that the items he proposes to furnish the Government under this specification are jointly approved by the National Institute for Occupational Safety and Health (NIOSH) and the Mining Enforcement and Safety Administration (MESA), U.S. Department of the Interior, under 30 CFR 11, subpart H. The certification shall include the NIOSH approval number.

3.2 Compliance with the above requirements does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of his product.

3.3 Design. The apparatus shall be an individual respiratory protective device designed to provide the user with a respirable atmosphere from compressed gas cylinder(s). The apparatus shall consist of three major groups: (1) the facepiece group, (2) the hose and regulator group, and (3) the backpack and harness group. A suitcase type carrying case (see 3.5.7) shall be furnished with each apparatus for transportation and storage purposes.

3.4 Material. All apparatus furnished under this specification shall be of the manufacturer's most recent production model approved in accordance with 30 CFR 11 subpart D. The materials used for the manufacture and assembly of the apparatus shall be new.

3.5 Components. All like components used in the assembly of the breathing apparatus produced under the responsibility of a prime manufacturer (contractor) shall be interchangeable.

3.5.1 Facepiece. The facepiece shall be of the design that covers the entire face and shall be so constructed as to assure a quick gas-tight fit on persons of varying facial shapes. A means shall be provided so that the facepiece may be suspended on the user in a convenient-to-use location when the apparatus is donned but not in use.

3.5.1.1 Lenses. The facepiece shall be equipped with a single lens. Lens replacement shall be readily accomplished in the field.

3.5.2 Breathing tube assembly. The breathing tube assembly shall be a corrugated flexible hose molded from an elastomer or other nonmetallic material. The length shall be sufficient to connect the facepiece to the regulator so that when worn, free movement of the head in any direction will be permitted without kinking the hose or placing stress on the hose connections.

3.5.3 Compressed gas cylinders. Compressed gas cylinders shall conform to 30 CFR 11 subpart H.

3.5.3.1 Military agencies. In addition to the requirements of 3.5.3, cylinders for Military agencies shall be color coded for cylinder identification and titled to show the exact identification of the material contained in accordance with the provisions of MIL-STD-101.

3.5.4 Harness assembly. The apparatus shall be provided with an adjustable fit harness for comfortably supporting the apparatus securely on the wearer's back. The harness shall include suitable means to prevent shifting of the apparatus on the wearer's back, during physical work. The shoulder strap or connection shall have a breaking strength of not less than 175 pounds (see 4.3.1).

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3.5.5 Audible warning device. The breathing apparatus shall be equipped with a warning device. The warning device shall sound an audible alarm when the remaining service life of the apparatus is reduced within a range of 20 to 25 percent of its rated service time.

3.5.6 High pressure hose leakage. The high pressure hose connecting the breathing air cylinder to the regulator shall not leak when tested as specified in 4.3.3.

3.5.7 Carrying case.

3.5.7.1 Design and construction. A carrying case shall be provided to retain the breathing apparatus, manuals, and wrench, if required. The case may be made of molded plastic, aluminum alloy, vulcanized fiber, or other suitable material. When aluminum alloy is used, it shall be anodized or treated with a chromate chemical conversion process and then shall be coated with one coat of zinc chromate primer and one coat of enamel. The case shall be provided with corrosion-resistant hinges and catches to insure closure, and a handle for ease of carrying. The case shall be of a design to provide compact enclosure of the apparatus and accessories without danger of deformation of any part due to closure. The case shall be provided with suitable built-in brackets, **bracing** , or straps for holding the apparatus securely in place under severe handling conditions. Sponge rubber or other suitable cushioning material shall be provided at points where shock stresses could be transmitted to the apparatus. The weight of the carrying case shall not exceed 25 pounds.

3.5.7.2 Manuals. One copy each of the manufacturer's standard operator and maintenance manual and repair parts list shall be furnished with each apparatus.

3.5.7.3 Wrench. When required by the manufacturer's design, a suitable wrench for connecting the high pressure hose to the compressed-gas cylinder and the regulator shall be furnished with each apparatus.

3.6 Workmanship. The apparatus shall be free from contamination (grease, oil, or other foreign matter) and damage (dents, scratches, or punctures) and other defects which might affect proper functioning.

3.6.1 Breathing tubes. The breathing tubes shall be free from tackiness, porous areas, crack, flashes, abrasions, and blisters. The elastomer shall be compact and free from bubbles and foreign matter. Ridges of mold parting lines, when closely trimmed, shall not be considered as flash.

3.6.2 Metal surfaces. The metal parts shall have no exterior sharp edges or burrs.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. Two samples of each class and style breathing apparatus assemblies shall be selected at random from the first lot produced for the tests specified in 4.3.1 thru 4.3.3. If a sample fails to comply with any of these tests, two additional samples shall be selected. If any additional sample fails any of the tests, breathing apparatus represented by the samples shall be considered as not conforming to the requirements of this specification. Manufacturers may correct minor deficiencies that cause failure. Lots subsequent to the first lot shall be tested only upon change in design, material, method of manufacture, or location of manufacturing facility. Sampling for visual examination (see 4.3.4) shall be 100 percent.

4.3 Test methods.

4.3.1 Strength of shoulder straps. The test shall be conducted in accordance with method 4108 or 5100 of Fed. Test Method Std. No. 191. When method 5100 is used, the distance between the clamps shall be 6 inches at the start of the test. All joints under test shall be midway between the clamps in the plane of pull.

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4.3.2 Audible alarm operation. The cylinder shall be charged to 40 percent or greater of its rated pressure. The cylinder valve shall be fully opened to pressurize the entire system. The by-pass value shall be opened slowly to vent the system pressure, observing the pressure turn-on of the audible alarm. The pressure at which the alarm starts to sound and ceases to sound shall be recorded, using a master gage of sufficient sensitivity. The pressure readings shall be within the limits specified in 3.5.5.

4.3.3 High pressure hose test. Connect the high pressure hose to a fully charged breathing air cylinder and regulator in accordance with the manufacturer's instructions. Close the regulator shut-off and by-pass values and fully open the cylinder valve. Apply soapy water to the hose and observe for conformance to 3.5.6. The appearance of bubbles indicates a leak.

4.3.4 Visual examination. The apparatus shall be visually examined for compliance with the requirements of 3.1, 3.5, and 3.6.

4.4 Inspection of preparation for delivery requirements. An inspection shall be made to determine whether the packaging, packing and marking requirements comply with section 5 of this specification. Defects shall be scored in accordance with table II. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The lot sizes shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 3.5 defects per hundred units.

TABLE II. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

5. PREPARATION

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Prior to preservation and packaging, the breathing apparatus and all components shall be thoroughly cleaned and dried in accordance with the applicable methods as specified in MIL-P-116.

5.1.1.2 Preservation and packaging. The complete apparatus with all components, when packed in the carrying case as specified in 3.5.7, shall be cushioned as required to prevent movement. The complete apparatus as packed in the carrying case shall be considered as a method III pack in accordance with MIL-P-116.

5.1.2 Level C. Cleaning, drying, preservation and packaging shall be in accordance with the manufacturer's standard commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Breathing apparatus, when preserved and packaged as specified in 5.1.1.2, shall be packed in class weather-resistant containers conforming to PPP-B-636. Closure, sealing, and strapping of containers shall be in accordance with the specification and appendix thereto.

5.2.2 Level B. Same as 5.2.1, except containers shall be class domestic and the closure of the boxes shall be in accordance with method II of the appendix to PPP-B-636.

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5.2.3 Level C. Breathing apparatus packaged in accordance with 5.1, which require overpacking for acceptance by the carrier, shall be packed in exterior type shipping containers in a manner that will assure safe transportation at the lowest rate to the point of delivery. Containers shall comply with the National Motor Freight Classification rules or the Uniform Freight Classification rules.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special markings required in the contract or order (see 6.2), shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special markings required in the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.3.2.1 Shipment by Military aircraft. When shipment is to be made by Military aircraft, the containers shall be marked in accordance with Manual AFM-71-4.

6. NOTES

6.1 Intended use. Breathing apparatus in accordance with this specification are designed for emergency entrance into a hazardous atmosphere. They are not for underground mine rescue or recovery operations except as auxiliary equipment.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Definitive part number (see 1.2.2).
(c) Levels of preservation and packaging and packing required (see 5.1 and 5.2).
(d) Whether civil or military marking is required (see 5.3).
(e) Special marking, if required (see 5.3).

MILITARY INTEREST:

Custodian:

Army - EA

Preparing activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITY:

HEW-NIH

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